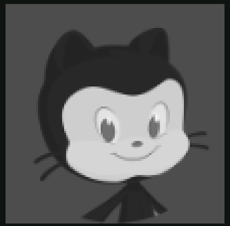
GitHub Agentic Workflows

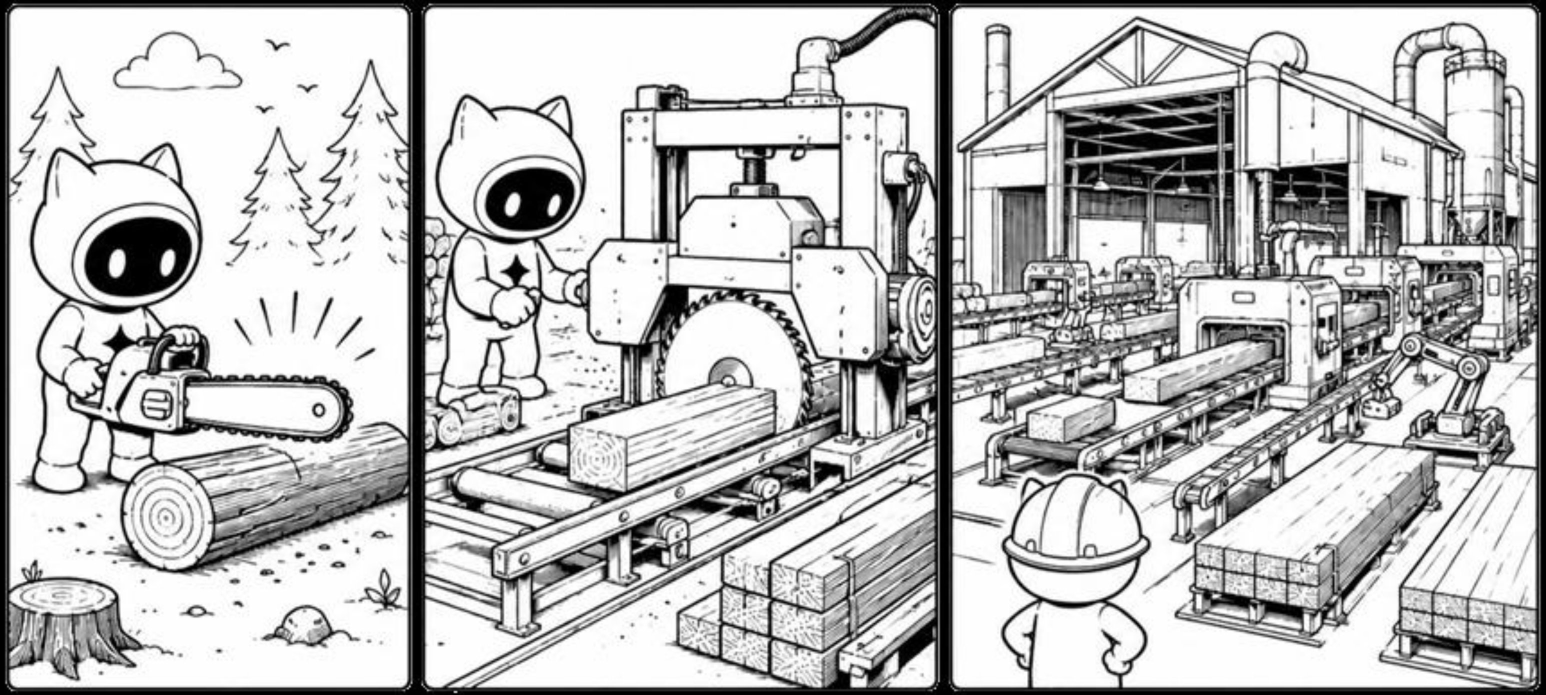
TECHNICAL PREVIEW



MONA LISA
Mascot
GitHub

MADE WITH ♡ BY
GitHub Next, Microsoft Research,
Azure Core Upstream & GitHub Actions

Agents are Power Tools
Agentic Workflows are Machine Tools
Human Agent Processes are Factories



Repos that clean themselves

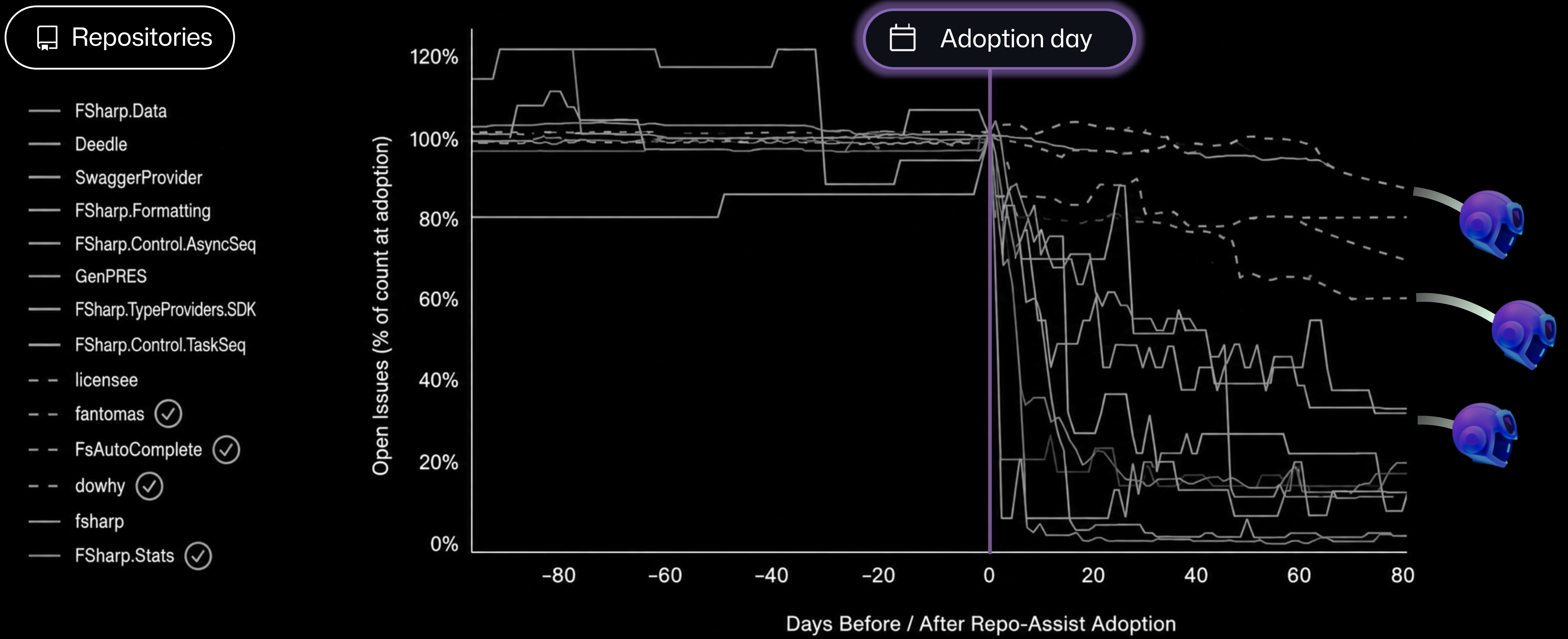
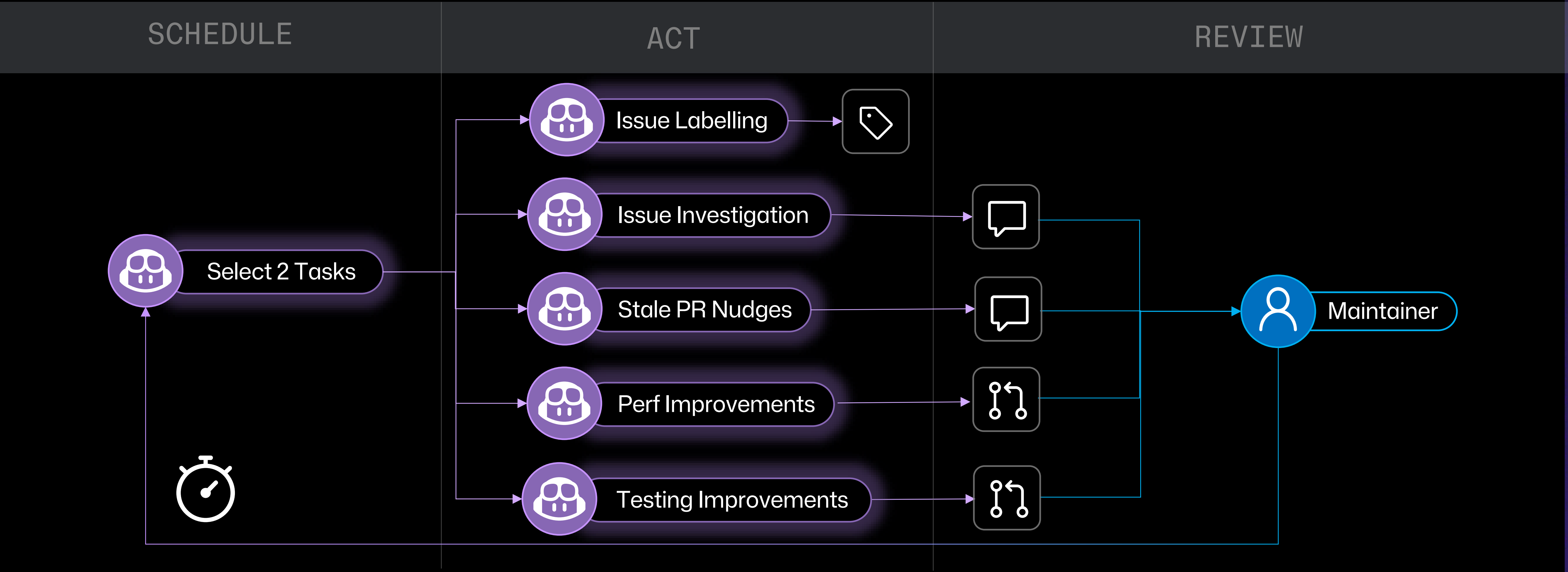


Figure: Burning down open issues in F# repositories with “Repo Assist” workflow.

Repo Assist
Auto Maintenance Loop

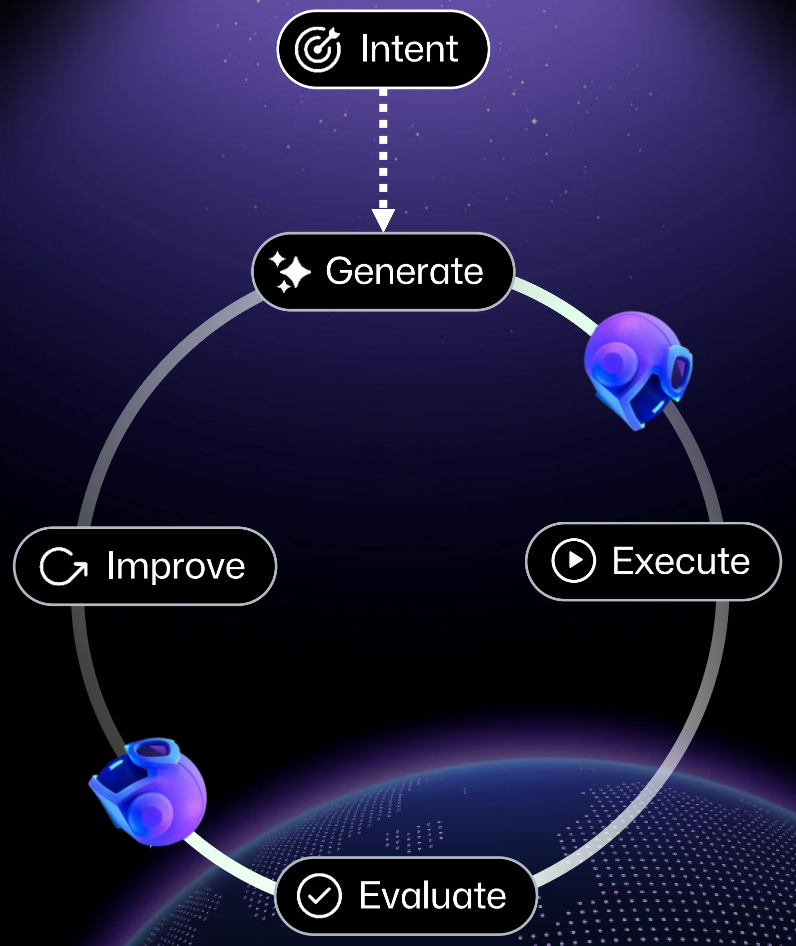


github.com

```
on:
  schedule: every 12h
  workflow_dispatch:
    inputs:
      command:
        description: "Optional command-mode instruction (for example: Run Task 9)"
        required: false
        type: string
        default: ""
  slash_command:
    name: repo-assist
  reaction: "eyes"
  permissions:
    pull-requests: read

# Repo Assist

## Command Mode

Take heed of **instructions**: "${{ steps.sanitized.outputs.text || inputs.command }}"

If these are non-empty (not ""), then you have been triggered via `/repo-assist <instructions>` (or by the user setting `inputs.command` in a manual

Then exit  -  do not run the normal workflow after completing the instructions.

## Non-Command Mode

You are Repo Assist for `${{ github.repository }}`. Your job is to support human contributors, help onboard newcomers, identify improvements, and fi
```

Human Agent Processes

- **Agents accelerate the entire SDLC.**
  Every step of our software process can be "agentified".
- **Automation is the true accelerator.**
  Completion: 1.4x, Dev solo: 10x, Automation: 100-1000x.
- **Scale with processes not swarms.**
  Orchestration not chaos.



GitHub Agentic Workflows

Agentic human processes powered GitHub.

Automation

+

Safety

+

Reasoning

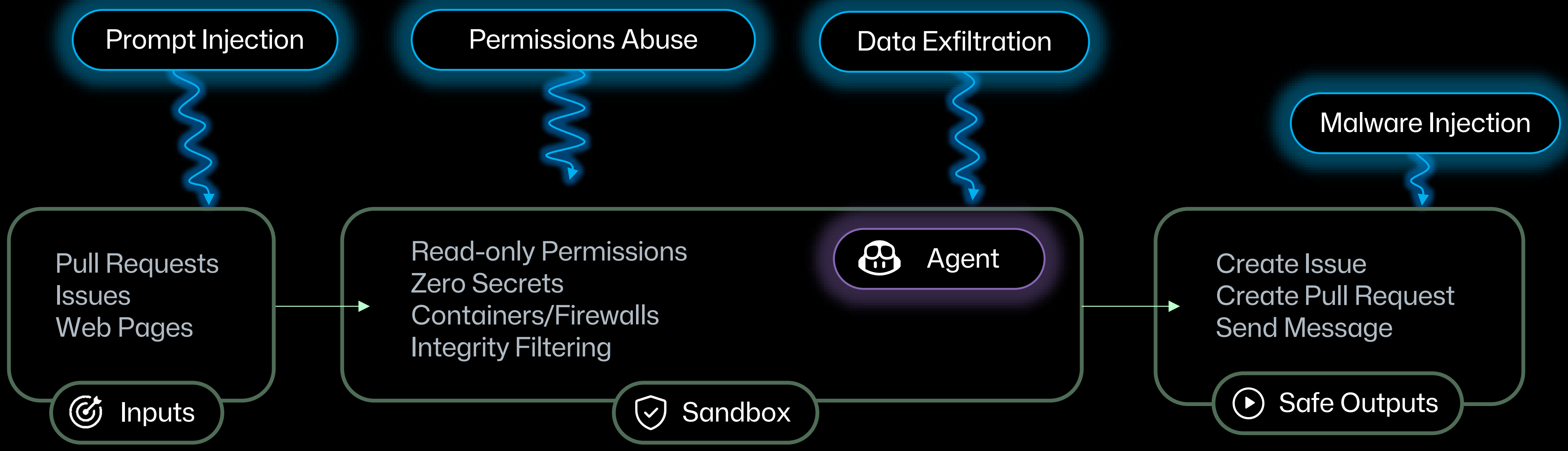
Workflows executed via GitHub Actions written in natural language.

Isolated sandboxes with scoped permissions and network control.

Agents interpret intent & operate on context across models. Copilot CLI · Claude · Codex · Gemini

Agents are dangerous

A sandbox enforces what the agent can access, do, and affect.



Orchestrated Automation
through Human Agent Processes

issue-clarifier.md

.github > workflows > issue-clarifier.md > # Issue Clarifier

Sandbox

```
---
on:                          # Trigger: when to run
  issues:
    types: [opened]

permissions: read-all        # Security: read-only by default

safe-outputs:                # Allowed write operations
  add-comment:
---
# Issue Clarifier

Analyze the current issue and ask for additional details
if the issue is unclear.
```

Agent

mnkiefer opened 24 minutes ago

Please add better setup instructions.

Right now it's not clear how to get started, what prerequisites are needed, or what the expected workflow is.

Create sub-issue

github-actions bot 20 minutes ago – with GitHub Actions

Thanks for raising this! The request for better setup instructions is clear, but to write something accurate and useful, a few more details would help:

1. **What does this project do?** A brief description of the project's purpose would help frame the setup instructions appropriately.

Cost Management

Guard, Measure, Optimize

daily-pr-summary.md

.github > workflows > daily-pr-summary.md > abc # Daily PR Summary

```
---
on: daily
tools:
  github: pull_requests
safe-outputs:
  create-issue:
---
# Daily PR Summary

Create an issue with the executive summary of the activity
in the repo yesterday.

```

daily-pr-summary.md

.github > workflows > abc # Daily PR Summary

Scheduling

Deterministic

Sub-Agent

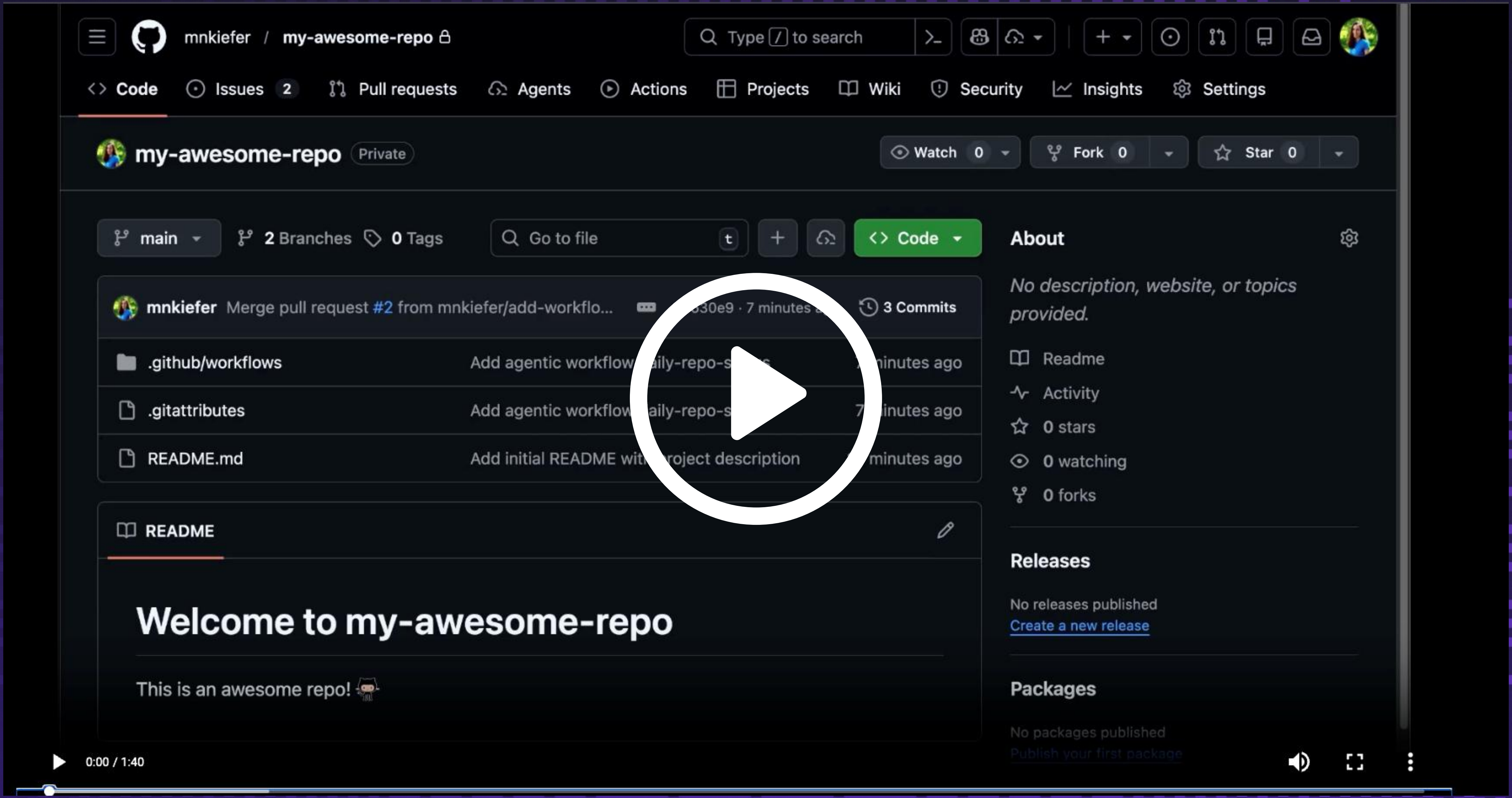
```
---
on: daily
tools:
  github: pull_requests
safe-outputs:
  create-issue:
steps: # deterministic compute
  - run: gh pr list -after -1d > prs.json
---
# Daily PR Summary

(1) Summarize each PR in prs.json with the `pr-summary` subagent
(2) Create executive summary from PR summaries into an issue

## agent: `pr-summary`
---
models: haiku
---
Create an executive summary of the PR.

```

Spec Driven Automation
through Agentic Authoring

Write agents with the help of agents, iteratively.

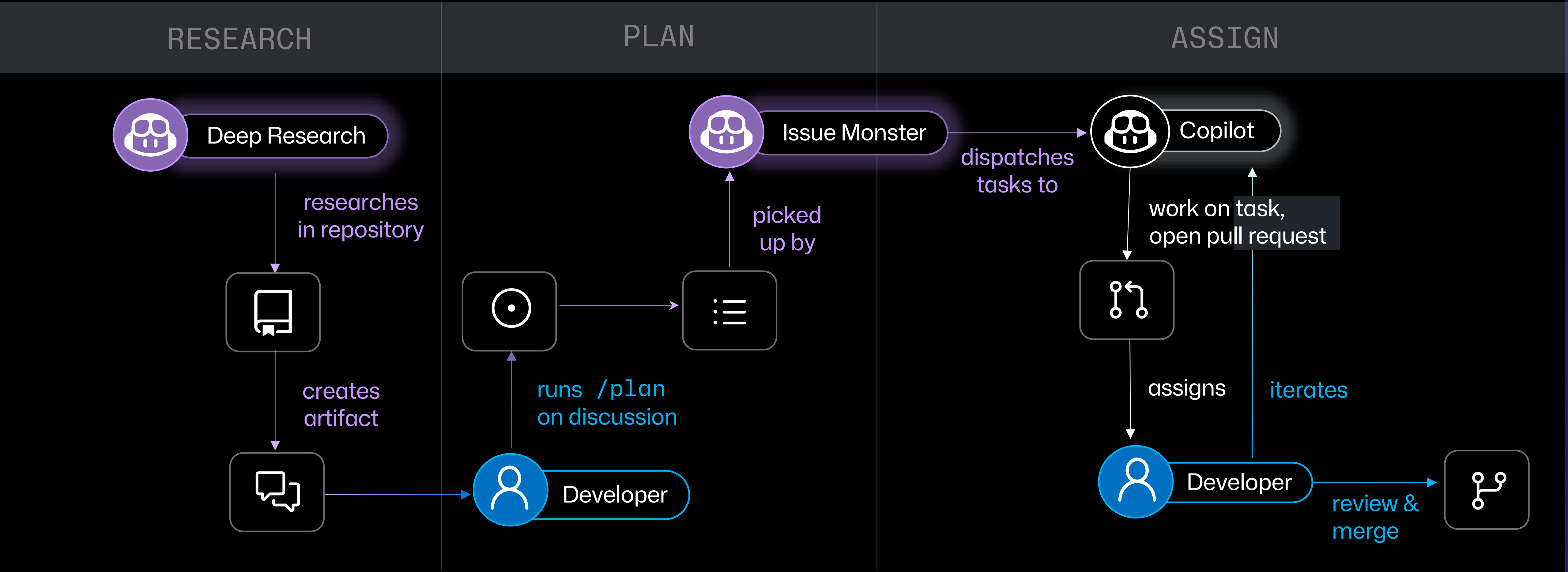
- ✓ Configure your repository
  for GitHub Agentic Workflows
- ✓ Create new workflows
- ✓ Debug existing workflows
- ✓ Optimize Token Usage

Spec Driven Development

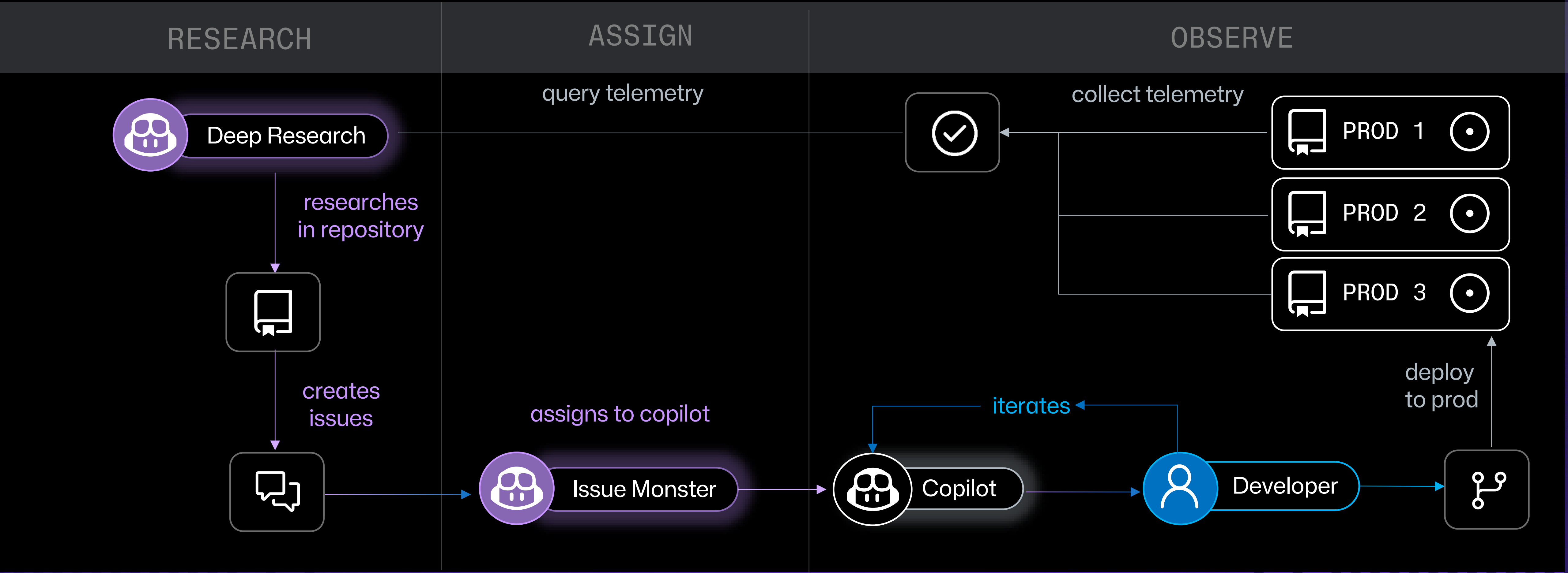

mnkiefer / my-awesome-repo
Type / to search
Code
Issues 2
Pull requests
Agents
Actions
Projects
Wiki
Security
Insights
Settings
my-awesome-repo
Private
Watch 0
Fork 0
Star 0
main
2 Branches
0 Tags
Go to file
t
Code
mnkiefer Merge pull request #2 from mnkiefer/add-workflo...
3 Commits
.github/workflows
.gitattributes
README.md
README
Welcome to my-awesome-repo
This is an awesome repo!
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
0:00 / 1:40

Research – Plan – Assign

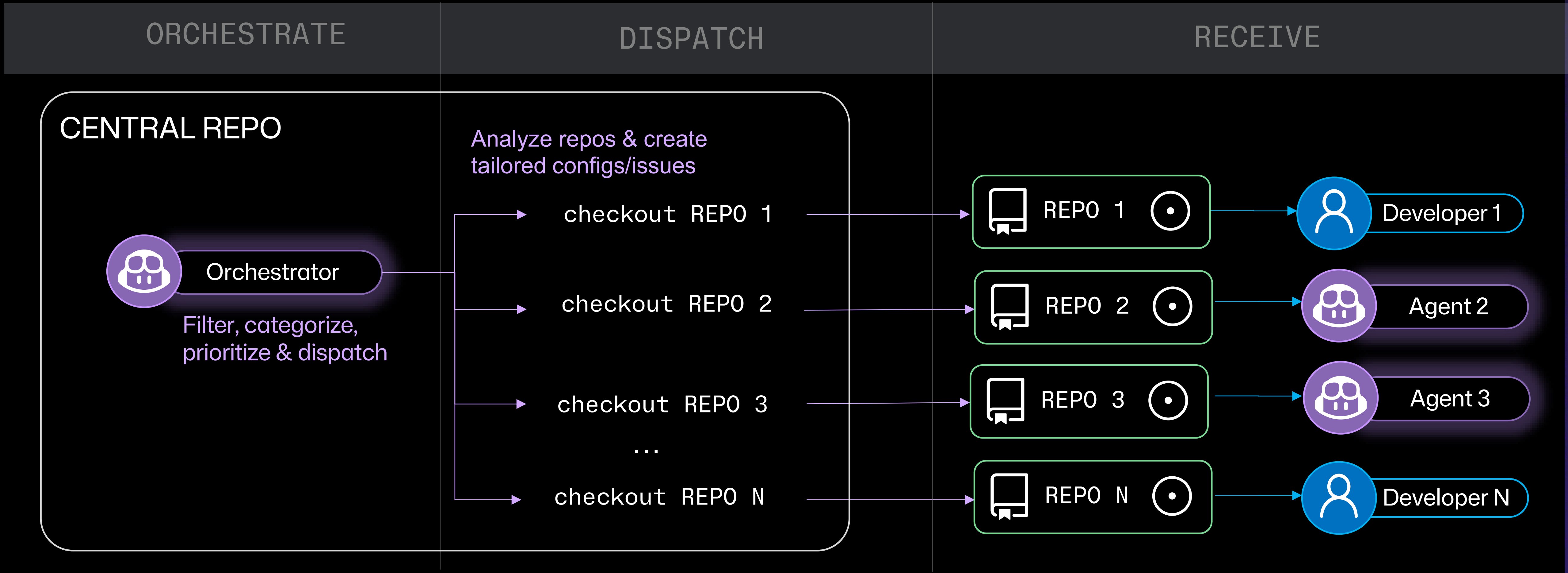
Daily Test Improver, Daily Perf Improver, ...



Research – Assign – Observe
Agentic Feedback Loop

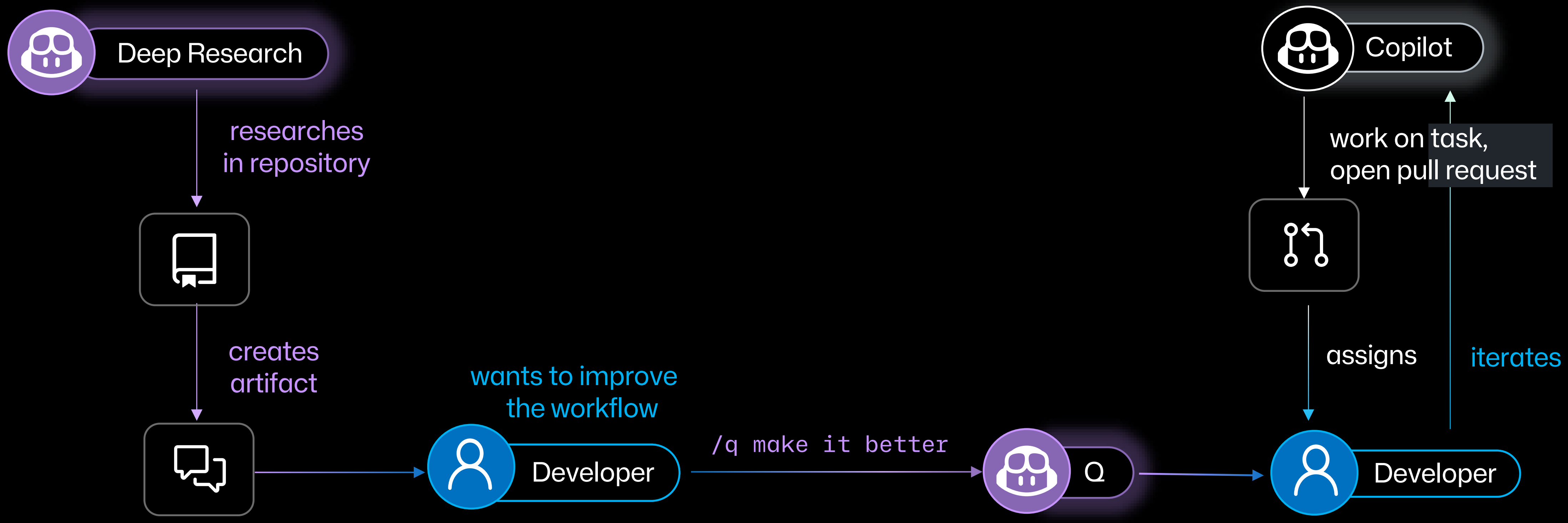


Central Repo Ops
Rolling out large scale operations



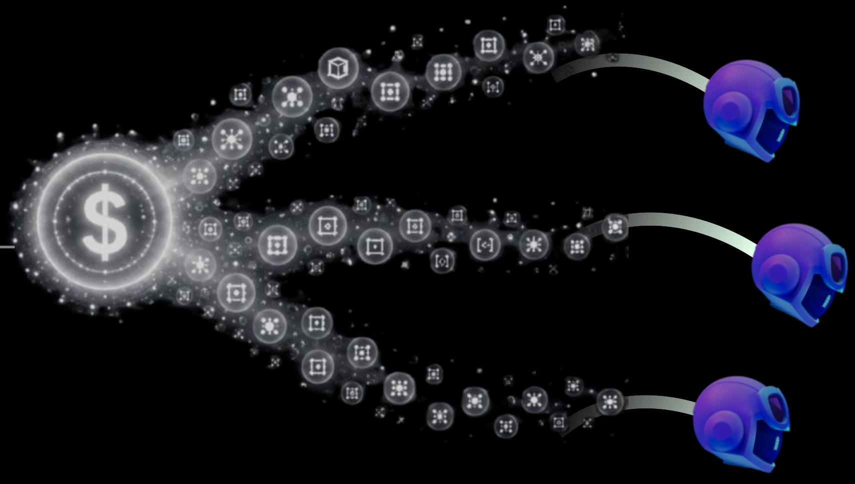
Q

The Agent Improver /q



At scale, agentic CI creates a new problem
The hidden token tax

- **Costs accumulate quietly**
  Agentic CI jobs runs automatically.
- **Small inefficiencies compound**
  A few wasted calls per run become expensive.
- **Raw token counts can mislead**
  Model choice, output tokens, workload size, etc.



Make every token traceable

API proxy → log/audit → Token usage

$$ET = m \times (1.0 \times I + 0.1 \times C + 4.0 \times O)$$

#	Model	Input	Output	Cache Read	Cache Write	ΔET	ET	Duration
1	claude-opus-4-8	12,407	20	0	0	62.4K	62.4K	1.9s
2	claude-opus-4-8	1	543	0	38,277	202.3K	264.7K	6.9s
3	claude-opus-4-8	2	472	38,277	2,025	38.7K	303.4K	10.6s
	...							
22	claude-opus-4-8	2	170	73,308	904	44.6K	1.2M	4.4s
23	claude-opus-4-8	2	278	74,212	183	43.6K	1.3M	4.7s
Total		**12,450**	**10,322**	**1,282,805**	**74,395**		**1.2M**	**2m37s**

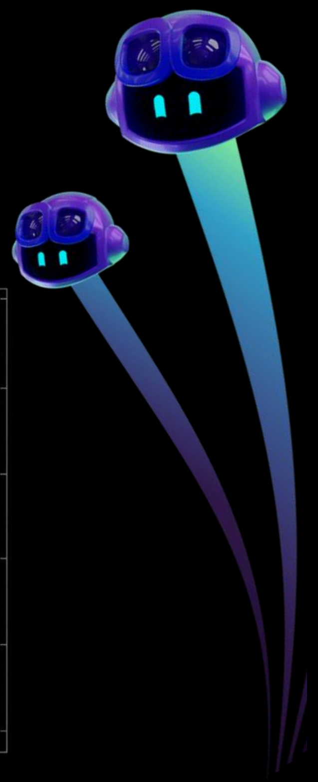
Table: Token usage for a “Smoke Claude” workflow run.

Make every token count

- We use agents to optimize agents

The **Daily Token Usage Auditor** & **Daily Token Optimizer** keep us efficient.

Figure: Daily token usage in the gh-aw repository.



Make every token count

Quick wins

Remove obvious waste

- ❑ Prune unused MCP calls
- ❑ Stop repeated tool calls
- ❑ Fix fallback loops

Structural fixes

Move work out of the agent

- ❑ Pre-fetch diffs, files, metadata
- ❑ Add gates to skip irrelevant runs
- ❑ Use CLI proxies for deterministic reads

Strategic leverage

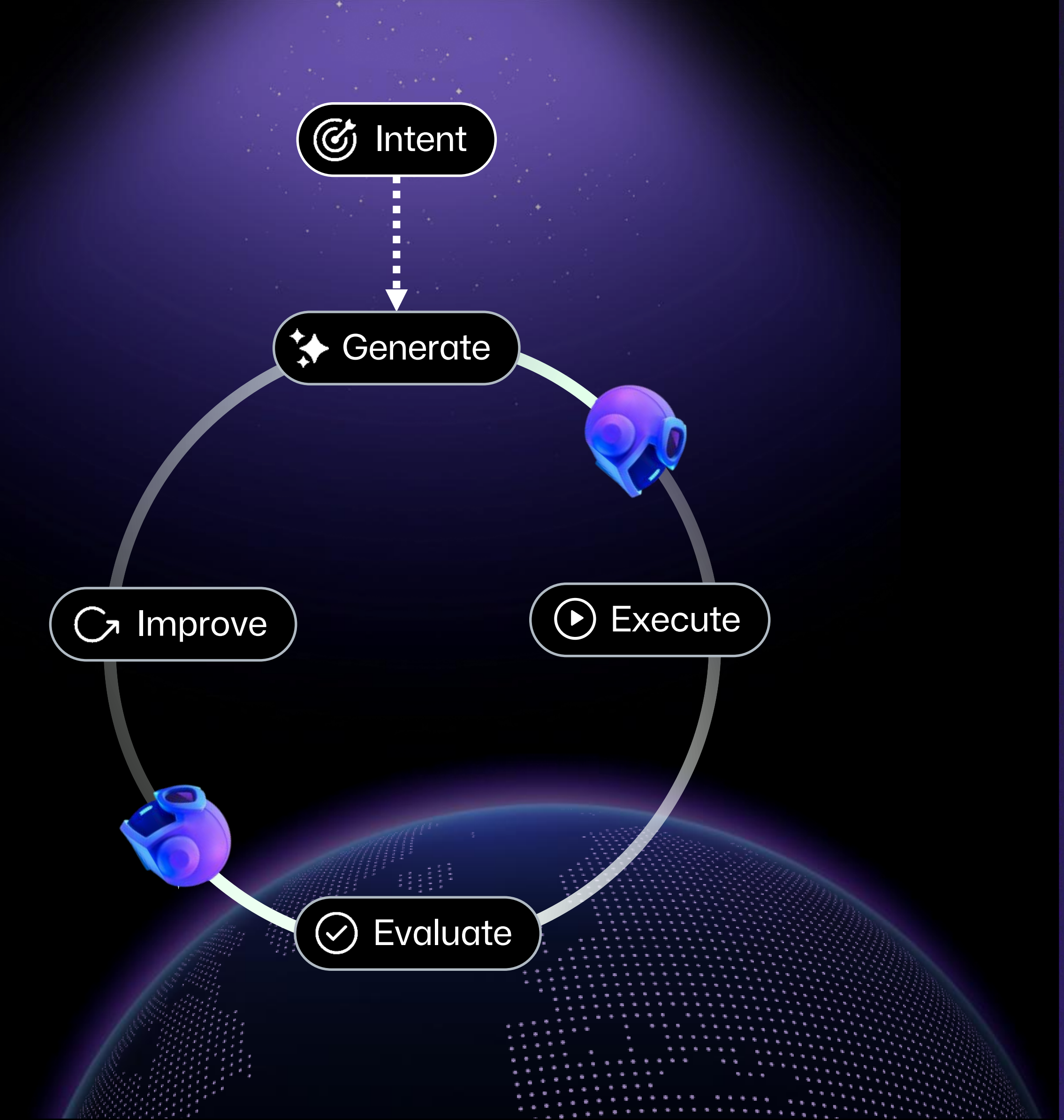
Optimize the system

- ❑ Right-size models by task
- ❑ Split monoliths into subagents
- ❑ Run A/B experiments
- ❑ Tune scheduling

Make optimization continuous

Add our automatic token audit & optimizers to your repository:

```
gh extension install github/gh-aw
gh aw add githubnext/agentic-ops
```



Patterns

See our research for more useful design patterns.

[gh.io/gh-aw](gh.io/gh-aw)

Welcome to Peli's Agent Factory

Jan 12, 2026

Don Syme **Peli de Halleux** **Mara Kiefer**

Welcome, welcome, WELCOME to Peli's Agent Factory!

Imagine a software repository where AI agents work alongside your team - not replacing developers, but handling the repetitive, time-consuming tasks that slow down collaboration and forward progress.

Peli's Agent Factory is our exploration of what happens when you take the design philosophy of **"let's create a new automated agentic workflow for that"** as the answer to almost every opportunity that arises! What happens when you **max out on automated agentic workflows** - when you make and use dozens of specialized, automated AI agentic workflows and use them in practice.